

Boosted Gaussian Witness Kick – A Manufactured Solution

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June 10, 2026

1 Purpose

This note defines a manufactured solution for a low-energy witness e^-/e^+ pair, kicked by the electromagnetic field of two highly relativistic colliding Gaussian electron bunches. The goal is to guide an OPALX implementation and establish an order-of-magnitude estimate.

It is understood that for the detector design, of a $\gamma\text{-}\gamma$ collider, a full self-consistent model is essential.

The terminology used below is:

- *colliding Gaussian beams*: the two prescribed high-energy electron bunches representing the beams at the interaction point;
- *source*: a field-producing bunch. In the present OPALX witness implementation this is particle container 0, which defines the mesh and field solve. In the manufactured model the source index j labels one of the prescribed colliding Gaussian bunches;
- *witness*: the low-energy e^-/e^+ particles. They sample the source fields but do not deposit charge in the present model.

Each source Gaussian bunch is prescribed as a uniformly, non-accelerated moving spherical Gaussian charge distribution in its own rest frame. The rest-frame Coulomb field is known analytically and is Lorentz-transformed to the lab frame at the witness event. The resulting lab-frame field is then applied to the witness particles with the same Boris kick convention used by OPALX.

2 Current OPALX Witness Model

The present OPALX witness treatment is a source-frame field sampling model. The source bunch defines the mesh and the field solve. A witness particle is not deposited on that mesh; instead its position is translated and rotated into the source beam-beam frame, the already available mesh field is gathered there, and the gathered vectors are rotated back before applying the usual particle kick.

In this approximation the witness is represented only by its spatial coordinate in the active tracking step. The relevant mapping is geometric,

$$\mathbf{r}_s = R_s (\mathbf{r}_w + (s_w - s_s)\hat{\mathbf{z}}), \quad (2.1)$$

where s_s and s_w are the source and witness path lengths and R_s is the rigid rotation from the tracking reference coordinates into the local source-beam coordinates used by the **BeamBeam** field mesh. It is an orthonormal 3×3 spatial rotation, not a Lorentz boost. In the straight coordinate

system used by the present tests, with no crossing angle and no curved reference trajectory, the two frames are aligned and $R_s = I$. It is kept explicitly because the implementation also has to support a rotated source reference frame. The sampled fields are then

$$\mathbf{E}_{\text{used}} = R_s^T \mathbf{E}_s(\mathbf{r}_s), \quad \mathbf{B}_{\text{used}} = R_s^T \mathbf{B}_s(\mathbf{r}_s). \quad (2.2)$$

This is useful as a first implementation because it reuses the **BeamBeam** mesh, but it does not treat the field evaluation as a Lorentz-covariant event calculation.

The full relativistic reference model will eventually start from the witness event

$$x_w^\mu = (ct_w, \mathbf{x}_w) \quad (2.3)$$

and evaluates the contribution of each source bunch j at that event. Let the source Gaussian bunch move with almost constant **BeamBeam** velocity

$$\mathbf{v}_j = c\boldsymbol{\beta}_j, \quad \gamma_j = \frac{1}{\sqrt{1 - \boldsymbol{\beta}_j^2}}, \quad (2.4)$$

and define the event relative to the source centroid by

$$\Delta t_j = t_w - t_{j0}, \quad \Delta \mathbf{x}_j = \mathbf{x}_w - \mathbf{X}_{j0}. \quad (2.5)$$

With the decomposition into components parallel and transverse to $\boldsymbol{\beta}_j$, the Lorentz transformation into the source rest frame is

$$ct'_j = \gamma_j (c\Delta t_j - \boldsymbol{\beta}_j \cdot \Delta \mathbf{x}_j), \quad (2.6)$$

$$\Delta \mathbf{x}'_{j,\parallel} = \gamma_j (\Delta \mathbf{x}_{j,\parallel} - c\boldsymbol{\beta}_j \Delta t_j), \quad \Delta \mathbf{x}'_{j,\perp} = \Delta \mathbf{x}_{j,\perp}. \quad (2.7)$$

The term $\boldsymbol{\beta}_j \cdot \Delta \mathbf{x}_j / c$ is the relativity-of-simultaneity piece that is absent in the current geometric sampling model.

For the manufactured Gaussian reference used here, the source rest-frame charge density is

$$\rho'_j(r') = \frac{Q_j}{(2\pi)^{3/2} \sigma_j^3} \exp\left(-\frac{r'^2}{2\sigma_j^2}\right), \quad (2.8)$$

and the electrostatic field is

$$\mathbf{E}'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\mathbf{r}'}{r'^3} \left[\text{erf}\left(\frac{r'}{\sqrt{2}\sigma_j}\right) - \sqrt{\frac{2}{\pi}} \frac{r'}{\sigma_j} \exp\left(-\frac{r'^2}{2\sigma_j^2}\right) \right], \quad \mathbf{B}'_j = 0. \quad (2.9)$$

More generally, the source rest-frame field tensor is transformed back to the lab/reference frame,

$$F_{j,\text{lab}}^{\mu\nu}(x_w) = \Lambda^\mu{}_\alpha(\boldsymbol{\beta}_j) F_j^{\prime\alpha\beta}(x'_j) \Lambda^\nu{}_\beta(\boldsymbol{\beta}_j). \quad (2.10)$$

The tensor expression above is the compact matrix form. For comparison with the code paths below, the same transformation is kept in component form as

$$\mathbf{E}_j = \gamma_j (\mathbf{E}'_j - \mathbf{v}_j \times \mathbf{B}'_j) - \frac{\gamma_j^2}{\gamma_j + 1} \frac{\mathbf{v}_j}{c^2} (\mathbf{v}_j \cdot \mathbf{E}'_j), \quad (2.11)$$

$$\mathbf{B}_j = \gamma_j \left(\mathbf{B}'_j + \frac{\mathbf{v}_j \times \mathbf{E}'_j}{c^2} \right) - \frac{\gamma_j^2}{\gamma_j + 1} \frac{\mathbf{v}_j}{c^2} (\mathbf{v}_j \cdot \mathbf{B}'_j). \quad (2.12)$$

The fields acting on the witness are the superposition

$$\mathbf{E}(x_w) = \sum_j \mathbf{E}_j(x_w), \quad \mathbf{B}(x_w) = \sum_j \mathbf{B}_j(x_w). \quad (2.13)$$

For the OPALX normalized momentum $\mathbf{P} = \gamma_w \boldsymbol{\beta}_w$, the corresponding witness equations of motion are

$$\frac{d\mathbf{x}_w}{dt} = c \frac{\mathbf{P}}{\sqrt{1 + \mathbf{P}^2}}, \quad \frac{d\mathbf{P}}{dt} = \frac{q}{mc} \left[\mathbf{E}(x_w) + c \frac{\mathbf{P}}{\sqrt{1 + \mathbf{P}^2}} \times \mathbf{B}(x_w) \right]. \quad (2.14)$$

This is the model the manufactured solution is meant to test against: event transformation into each source rest frame, field evaluation there, field-tensor transformation back to the lab/reference frame, and then the witness kick.

During an active `BEAMBEAM` window, OPALX treats particle container 0 as the source bunch. Additional containers listed through `WITNESS_CONTAINERS` are passive: their charge is not deposited onto the source mesh, and they only sample the field that was already computed from container 0.

Let the source container path length be s_s , the witness container path length be s_w , and the witness particle coordinate in its own container frame be

$$\mathbf{r}_w = (x_w, y_w, z_w). \quad (2.15)$$

The implemented longitudinal mapping into the source frame is

$$\Delta s = s_w - s_s, \quad z_s = z_w + \Delta s. \quad (2.16)$$

With the same R_s defined above, the sampled source-frame coordinate is

$$\mathbf{r}_s = R_s (\mathbf{r}_w + \Delta s \hat{\mathbf{z}}). \quad (2.17)$$

OPALX then gathers the current source mesh fields at this point:

$$\mathbf{E}_s(\mathbf{r}_s), \quad \mathbf{B}_s(\mathbf{r}_s). \quad (2.18)$$

After gathering, the witness particle positions are restored and the fields are rotated back with the inverse rigid rotation,

$$\mathbf{E}_{\text{used}} = R_s^T \mathbf{E}_s, \quad \mathbf{B}_{\text{used}} = R_s^T \mathbf{B}_s. \quad (2.19)$$

Those fields are passed to the normal per-container Boris kick. In OPALX units the particle momentum is

$$\mathbf{P} = \boldsymbol{\beta}_w \gamma_w, \quad (2.20)$$

and the continuous Lorentz-force equation represented by the kick is

$$\frac{d\mathbf{P}}{dt} = \frac{qc}{mc^2} \mathbf{E}_{\text{used}} + \frac{qc^2}{mc^2} \boldsymbol{\beta}_w \times \mathbf{B}_{\text{used}}. \quad (2.21)$$

2.1 Source Field Currently Available in OPALX

The `BeamBeam` field calculation uses the binned quasi-static field path. This statement includes the case where the adaptive binning has only one current bin: as long as a `BINNING` definition is attached to the field solver, OPALX dispatches to `computeBinnedSelfFields()` and applies the same quasi-static Lorentz correction to that single bin.

For a source bin b , OPALX computes the global mean normalized momentum

$$\bar{\mathbf{P}}_b = \langle \boldsymbol{\beta} \gamma \rangle_b, \quad (2.22)$$

and the bin Lorentz factor and mean velocity

$$\gamma_b = \sqrt{1 + \bar{\mathbf{P}}_b^2}, \quad \mathbf{v}_b = c \frac{\bar{\mathbf{P}}_b}{\gamma_b}. \quad (2.23)$$

Equivalently, the bin normalized velocity and its unit direction are

$$\boldsymbol{\beta}_b = \frac{\mathbf{v}_b}{c} = \frac{\bar{\mathbf{P}}_b}{\gamma_b}, \quad \hat{\boldsymbol{\beta}}_b = \frac{\boldsymbol{\beta}_b}{|\boldsymbol{\beta}_b|}. \quad (2.24)$$

Before the Poisson solve, the mesh density is transformed to the approximate bin rest frame by

$$\rho'_b = \frac{\rho_b}{\gamma_b}, \quad (2.25)$$

and the longitudinal mesh spacing is stretched by

$$\Delta z' = \gamma_b \Delta z. \quad (2.26)$$

After solving the electrostatic problem in this stretched frame, the code accumulates the lab-frame fields

$$\mathbf{E}_b = \gamma_b \mathbf{E}'_b - (\gamma_b - 1)(\mathbf{E}'_b \cdot \hat{\boldsymbol{\beta}}_b) \hat{\boldsymbol{\beta}}_b, \quad (2.27)$$

$$\mathbf{B}_b = \frac{\gamma_b}{c} \boldsymbol{\beta}_b \times \mathbf{E}'_b. \quad (2.28)$$

The electric-field transformation could equivalently be written in matrix form as

$$\mathbf{E}_b = \left[\gamma_b I - (\gamma_b - 1) \hat{\boldsymbol{\beta}}_b \hat{\boldsymbol{\beta}}_b^T \right] \mathbf{E}'_b, \quad (2.29)$$

and the magnetic field can be written in the same operator style by introducing the skew-symmetric cross-product matrix

$$[\boldsymbol{\beta}_b]_{\times} = \begin{pmatrix} 0 & -\beta_{b,z} & \beta_{b,y} \\ \beta_{b,z} & 0 & -\beta_{b,x} \\ -\beta_{b,y} & \beta_{b,x} & 0 \end{pmatrix}, \quad [\boldsymbol{\beta}_b]_{\times} \mathbf{a} = \boldsymbol{\beta}_b \times \mathbf{a}. \quad (2.30)$$

Then Eq. (2.28) is equivalently

$$\mathbf{B}_b = \frac{\gamma_b}{c} [\boldsymbol{\beta}_b]_{\times} \mathbf{E}'_b. \quad (2.31)$$

The component forms in Eqs. (2.27) and (2.28) are still kept because they make the parallel and transverse parts visible and mirror the implementation. The accumulated fields are then gathered to source particles, and the current mesh fields can also be sampled by witness containers.

2.2 What Is Missing

The witness model is still a source-frame field sampling approximation. It rotates and translates witness geometry into the source beam frame, samples the mesh, and rotates the sampled vectors back. It does not evaluate the electromagnetic field tensor at the witness event.

A covariant treatment would start from a witness event

$$x_w^\mu = (ct_w, \mathbf{x}_w) \quad (2.32)$$

and transform that event into the rest frame of each source bunch. For a source with velocity $\mathbf{v}_s = c\boldsymbol{\beta}_s$, the parallel part of the event transform is

$$t'_s = \gamma_s \left(t_w - \frac{\boldsymbol{\beta}_s \cdot \mathbf{x}_w}{c} \right), \quad (2.33)$$

$$\mathbf{x}'_{s,\parallel} = \gamma_s (\mathbf{x}_{w,\parallel} - \mathbf{v}_s t_w), \quad \mathbf{x}'_{s,\perp} = \mathbf{x}_{w,\perp}. \quad (2.34)$$

The field tensor would then be transformed back to the lab/reference frame,

$$F_{\text{lab}}^{\mu\nu} = \Lambda^\mu{}_\alpha F_s'^{\alpha\beta} \Lambda^\nu{}_\beta. \quad (2.35)$$

In vector notation, for a source rest-frame field $\mathbf{E}'$, $\mathbf{B}'$, the lab field is

$$\mathbf{E} = \gamma_s (\mathbf{E}' - \mathbf{v}_s \times \mathbf{B}') - \frac{\gamma_s^2}{\gamma_s + 1} \frac{\mathbf{v}_s}{c^2} (\mathbf{v}_s \cdot \mathbf{E}'), \quad (2.36)$$

$$\mathbf{B} = \gamma_s \left(\mathbf{B}' + \frac{\mathbf{v}_s \times \mathbf{E}'}{c^2} \right) - \frac{\gamma_s^2}{\gamma_s + 1} \frac{\mathbf{v}_s}{c^2} (\mathbf{v}_s \cdot \mathbf{B}'). \quad (2.37)$$

The current witness gather does not perform this per-source or per-witness event transformation.

The missing physics and numerical pieces are:

- no Lorentz transformation of the witness event into each source frame;
- no relativity-of-simultaneity term in the sampled source coordinate;
- no full $F^{\mu\nu}$ transformation for the sampled field;
- no retarded-time or Lienard-Wiechert treatment for accelerated sources;
- no deposition of witness charge back onto the source mesh;
- no witness-witness or electron-positron mutual space charge;
- no separate Lorentz transformation of the already accumulated source mesh fields into the low-energy witness frame.

3 Manufactured Boosted-Gaussian Model

For a controlled implementation target, prescribe each source bunch j by

$$Q_j, \quad \sigma'_j, \quad \beta_j, \quad \gamma_j = \frac{1}{\sqrt{1 - \beta_j^2}}, \quad (3.1)$$

and by the lab-frame center worldline

$$\mathbf{R}_j(t) = \mathbf{R}_{j0} + \beta_j c(t - t_0). \quad (3.2)$$

At the witness event $(t, \mathbf{r}_w)$, define the lab-frame displacement from the source center,

$$\boldsymbol{\ell}_j = \mathbf{r}_w - \mathbf{R}_j(t). \quad (3.3)$$

Split it into parts parallel and perpendicular to the source velocity,

$$\ell_{j,\parallel} = (\boldsymbol{\ell}_j \cdot \hat{\boldsymbol{\beta}}_j) \hat{\boldsymbol{\beta}}_j, \quad \ell_{j,\perp} = \boldsymbol{\ell}_j - \ell_{j,\parallel}. \quad (3.4)$$

For a uniformly moving source, the corresponding source-rest coordinate at the same lab event is

$$\mathbf{r}'_j = \ell_{j,\perp} + \gamma_j \ell_{j,\parallel}. \quad (3.5)$$

The rest-frame charge density is a spherical Gaussian,

$$\rho'_j(\mathbf{r}') = \frac{Q_j}{(2\pi)^{3/2} \sigma_j'^3} \exp\left(-\frac{r'^2}{2\sigma_j'^2}\right). \quad (3.6)$$

The analytic rest-frame electrostatic potential and electric field are

$$\phi'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\text{erf}(u)}{r'}, \quad u = \frac{r'}{\sqrt{2}\sigma_j'}, \quad (3.7)$$

$$\mathbf{E}'_j(\mathbf{r}') = \frac{Q_j}{4\pi\epsilon_0} \frac{\text{erf}(u) - \sqrt{2/\pi} (r'/\sigma_j') \exp[-r'^2/(2\sigma_j'^2)]}{r'^3} \mathbf{r}', \quad (3.8)$$

with the regular limit $\mathbf{E}'_j(0) = 0$. Since the source is electrostatic in its own rest frame,

$$\mathbf{B}'_j = 0. \quad (3.9)$$

Boosting this field into the lab gives the simpler component form

$$\mathbf{E}_{j,\parallel} = \mathbf{E}'_{j,\parallel}, \quad \mathbf{E}_{j,\perp} = \gamma_j \mathbf{E}'_{j,\perp}, \quad (3.10)$$

$$\mathbf{B}_j = \frac{1}{c^2} \mathbf{v}_j \times \mathbf{E}_j. \quad (3.11)$$

The total manufactured lab field at the witness event is

$$\mathbf{E}_w(t, \mathbf{r}_w) = \sum_j \mathbf{E}_j(t, \mathbf{r}_w), \quad \mathbf{B}_w(t, \mathbf{r}_w) = \sum_j \mathbf{B}_j(t, \mathbf{r}_w). \quad (3.12)$$

3.1 Witness Kick

The manufactured witness uses the same normalized momentum convention as OPALX,

$$\mathbf{P} = \beta_w \gamma_w. \quad (3.13)$$

The fields used by the kick are exactly the total manufactured lab-frame fields from Eq. (3.12), evaluated at the current witness event:

$$\mathbf{E}_w^n = \mathbf{E}_w(t^n, \mathbf{r}_w^n), \quad \mathbf{B}_w^n = \mathbf{B}_w(t^n, \mathbf{r}_w^n). \quad (3.14)$$

They are therefore not the source-frame mesh fields $\mathbf{E}_s(\mathbf{r}_s), \mathbf{B}_s(\mathbf{r}_s)$ used in the current OPALX sampling approximation. The Python prototype applies the same Boris kick sequence used in OPALX:

$$\mathbf{P}^- = \mathbf{P}^n + \frac{1}{2}\Delta t \frac{qc}{mc^2} \mathbf{E}_w^n, \quad (3.15)$$

$$\mathbf{t} = \frac{1}{2}\Delta t \frac{qc^2}{\gamma^- mc^2} \mathbf{B}_w^n, \quad \mathbf{s} = \frac{2\mathbf{t}}{1 + \mathbf{t} \cdot \mathbf{t}}, \quad (3.16)$$

$$\mathbf{P}' = \mathbf{P}^- + \mathbf{P}^- \times \mathbf{t}, \quad (3.17)$$

$$\mathbf{P}^+ = \mathbf{P}^- + \mathbf{P}' \times \mathbf{s}, \quad (3.18)$$

$$\mathbf{P}^{n+1} = \mathbf{P}^+ + \frac{1}{2}\Delta t \frac{qc}{mc^2} \mathbf{E}_w^n. \quad (3.19)$$

3.2 Initial Manufactured Cases

The first Python prototype is `sandbox/python/boosted_gaussian_witness.py`. The current numerical examples use the large $\gamma\gamma$ BeamBeam diagnostic geometry:

$$L_{\text{BB}} = 0.32 \text{ m}, \quad R_{\text{BB}} = 0.15 \text{ m}, \quad s_{\text{IP}} = 0.17 \text{ m}. \quad (3.20)$$

The manufactured model is written in local time measured from pair creation at the IP. In the corresponding OPALX large-cylinder input this event is placed at about 550.3 ps, with the primary source crossing the IP at about 567.1 ps. The delayed-retirement diagnostic keeps the primary source active until 1000 ps in the OPALX clock, while the manufactured solution below isolates the continuum field model from the OPALX source-retirement bookkeeping. The table below was generated with

$$\begin{aligned} K_s &= 245 \text{ MeV}, \\ \gamma_s &= 480.453039976, \\ \beta_s &= 0.999997833949, \\ Q &= -2.002721 \text{ nC}, \\ \sigma' &= 0.6 \text{ mm}, \\ \Delta t &= 1 \text{ ps}, \\ \mathbf{P}_w^0 &= (0, 0, 0). \end{aligned} \quad (3.21)$$

The two symmetric source centers are both at the IP at $t = 0$, moving in opposite z -directions. The witness charge is an electron charge.

case	$\mathbf{r}_w$ [mm]	$ \mathbf{E} $ [GV/m]	$ \mathbf{B} $ [T]	$\Delta\mathbf{P}$
symmetric center	(0.000, 0.000, 0.000)	0.000	0.000	(0.000, 0.000, 0.000)
symmetric off-axis	(0.500, 0.000, 0.000)	8.682	0.000	(5.093, 0.000, 0.000)
single boosted source	(0.500, 0.000, 0.000)	4.341	14.480	(1.573, 0.000, 1.237)
asymmetric collision	(0.500, 0.000, 0.000)	7.814	2.896	(4.537, 0.000, 0.462)

The central symmetric case is intentionally a null test:

$$\mathbf{E}(0) = 0, \quad \mathbf{B}(0) = 0, \quad \Delta\mathbf{P} = 0. \quad (3.22)$$

The off-axis symmetric case checks the transverse electric kick while the two magnetic fields cancel. The single-source and asymmetric cases give nonzero magnetic fields and therefore exercise the full $(\mathbf{E}, \mathbf{B})$ Boris path.

3.3 Back-to-Back Breit-Wheeler Pair Propagation

The Python prototype also contains a two-particle witness test intended to be a minimal model for a low-energy Breit-Wheeler pair born at the interaction point. The electron and positron are initialized at the same lab event,

$$\mathbf{r}_{e-}(0) = \mathbf{r}_{e+}(0) = \mathbf{0}, \quad (3.23)$$

with equal kinetic energy

$$K_{e-} = K_{e+} = 313 \text{ keV}. \quad (3.24)$$

The corresponding normalized momentum magnitude is

$$|\mathbf{P}| = \sqrt{\left(1 + \frac{K}{m_e c^2}\right)^2 - 1} = 1.26500561229, \quad \beta = 0.784487091427. \quad (3.25)$$

For the default test direction $\hat{\mathbf{u}} = \hat{\mathbf{x}}$,

$$\mathbf{P}_{e-}(0) = |\mathbf{P}|\hat{\mathbf{u}}, \quad \mathbf{P}_{e+}(0) = -|\mathbf{P}|\hat{\mathbf{u}}. \quad (3.26)$$

Three trajectories are integrated with identical initial conditions and timestep:

- **symmetric_collision**: the pair samples the Lorentz-transformed fields of two equal boosted Gaussian source bunches;
- **asymmetric_collision**: the pair samples the same left source and a right source with 80% of the charge;
- **no_fields**: the same pair is propagated freely with $\mathbf{E} = \mathbf{B} = 0$.

All cases are written into the same trajectory CSV with a **scenario** column.

For the physical-charge illustration

$$\begin{aligned} Q &= -2.002721 \text{ nC}, & \sigma' &= 0.6 \text{ mm}, \\ \Delta t_{\text{fine}} &= 0.001 \text{ ps} \quad (0 \leq t \leq 5 \text{ ps}), & \Delta t_{\text{tail}} &= 0.1 \text{ ps}, \\ & & T &= 1050 \text{ ps}, \end{aligned} \quad (3.27)$$

the exact creation point is a symmetry null:

$$\mathbf{E}(0, \mathbf{0}) = \mathbf{0}, \quad \mathbf{B}(0, \mathbf{0}) = \mathbf{0}. \quad (3.28)$$

In the symmetric collision with the default pure transverse launch, both particles remain in the $z = 0$ symmetry plane, so the magnetic fields from the two equal and opposite source bunches cancel along the trajectory. The asymmetric collision breaks this cancellation and samples a nonzero magnetic field even for the same initial pair direction. The generated kinematic summary is evaluated at $T = 1050 \text{ ps}$. At the default 313 keV kinetic energy the free transverse path length over this interval is about 246.9 mm, so the manufactured trajectory crosses the $R_{\text{BB}} = 150 \text{ mm}$ cylinder scale within the plotted time window:

case	particle	$\mathbf{r}_f$ [mm]	$\mathbf{P}_f$	K_f [keV]	$\Delta\mathbf{r}$ [μm]	ΔK [eV]
symmetric collision	e^-	(246.953943, 0.000000, 0.000000)	(1.265158, 0.000000, 0.000000)	313.061	(11.464, 0.000, 0.000)	61.224
symmetric collision	e^+	(-246.931013, 0.000000, 0.000000)	(-1.264853, 0.000000, 0.000000)	312.939	(11.466, 0.000, 0.000)	-61.220
asymmetric collision	e^-	(246.952796, 0.000000, -0.001461)	(1.265143, 0.000000, -0.000007)	313.055	(10.317, 0.000, -1.461)	55.102
asymmetric collision	e^+	(-246.932160, 0.000000, 0.001462)	(-1.264868, 0.000000, 0.000007)	312.945	(10.319, 0.000, 1.462)	-55.098

The field evaluation and the particle push in this manufactured model are performed in the lab/reference frame. Each source field is obtained by transforming the current witness event into the corresponding source rest frame, evaluating the spherical Gaussian Coulomb field there, and boosting the field back to the lab frame. The magnetic field is included in the Boris rotation through

$$\mathbf{B}_j = \frac{1}{c^2} \mathbf{v}_j \times \mathbf{E}_j. \quad (3.29)$$

The vanishing magnetic contribution in the default symmetric pair run is therefore not an omission. It is a symmetry result: the pair starts at the overlap event and is launched transversely in the $z = 0$ plane, where the magnetic fields of the two equal counter-propagating electron bunches cancel. The asymmetric case breaks this cancellation and gives a nonzero magnetic field.

An earlier uniform $\Delta t = 0.1$ ps pair run was useful as a geometry check, but it under-resolved the force transient at bunch overlap. The table above and Figure 2 therefore use a multirate integration: 0.001 ps through the first 5 ps, followed by 0.1 ps steps after the source overlap has passed. Since the manufactured source has $\gamma_s = 480.453$ and $\sigma' = 0.6$ mm, the lab-frame longitudinal source scale is

$$\sigma_z = \frac{\sigma'}{\gamma_s} \simeq 1.25 \mu\text{m}, \quad \frac{\sigma_z}{c} \simeq 4.16 \text{ fs} = 0.00416 \text{ ps}. \quad (3.30)$$

Thus a timestep comparable to a few femtoseconds is required to resolve the initial kick. A short 5 ps timestep check for the symmetric electron gives

Δt [ps]	Δx [μm]	ΔK [eV]
0.1	0.00434	5.20
0.01	0.04295	48.64
0.005	0.05191	58.68
0.002	0.05387	60.86
0.001	0.05415	61.16

The coarse 0.1 ps run therefore underestimated the early energy kick by about a factor of twelve. Production manufactured-solution comparisons should use a timestep near 10^{-3} ps during overlap, or an equivalent adaptive integration around the source-overlap event.

Figure 1 shows the magnitude of the initial manufactured electric field for the symmetric and asymmetric source configurations. Figure 2 shows the resulting pair trajectories and their deviation from free propagation.

4 Implementation Use in OPALX

The recommended implementation sequence is:

1. Keep the Python evaluator as the reference for exact one-witness values.
2. Add an OPALX unit or regression test with one prescribed source and one witness particle.
3. First compare the field sampled by OPALX against the Python $(\mathbf{E}, \mathbf{B})$ value at the same event.
4. Then compare the witness momentum after one kick against the Python Boris update.

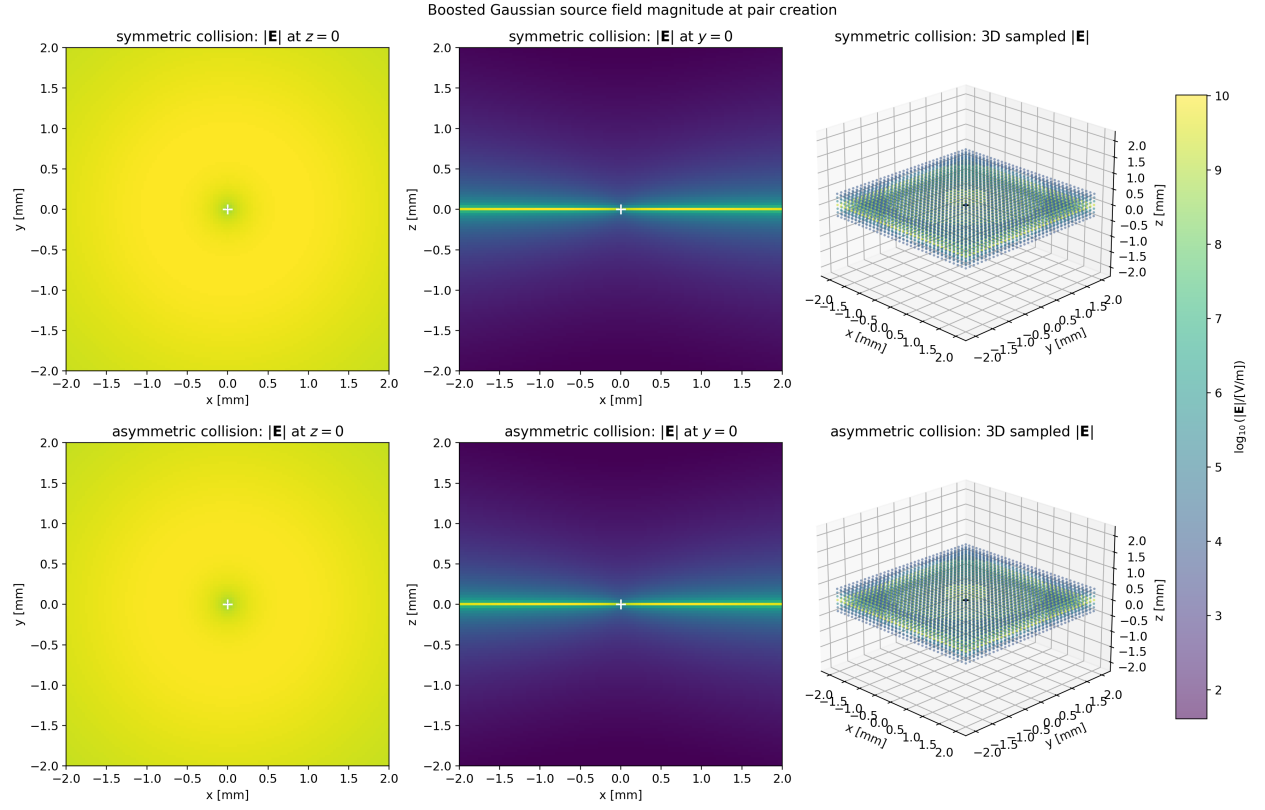


Figure 1: Magnitude of the manufactured boosted-Gaussian electric field at pair creation for the updated $\sigma' = 0.6$ mm source bunch. The white marker is the interaction point.

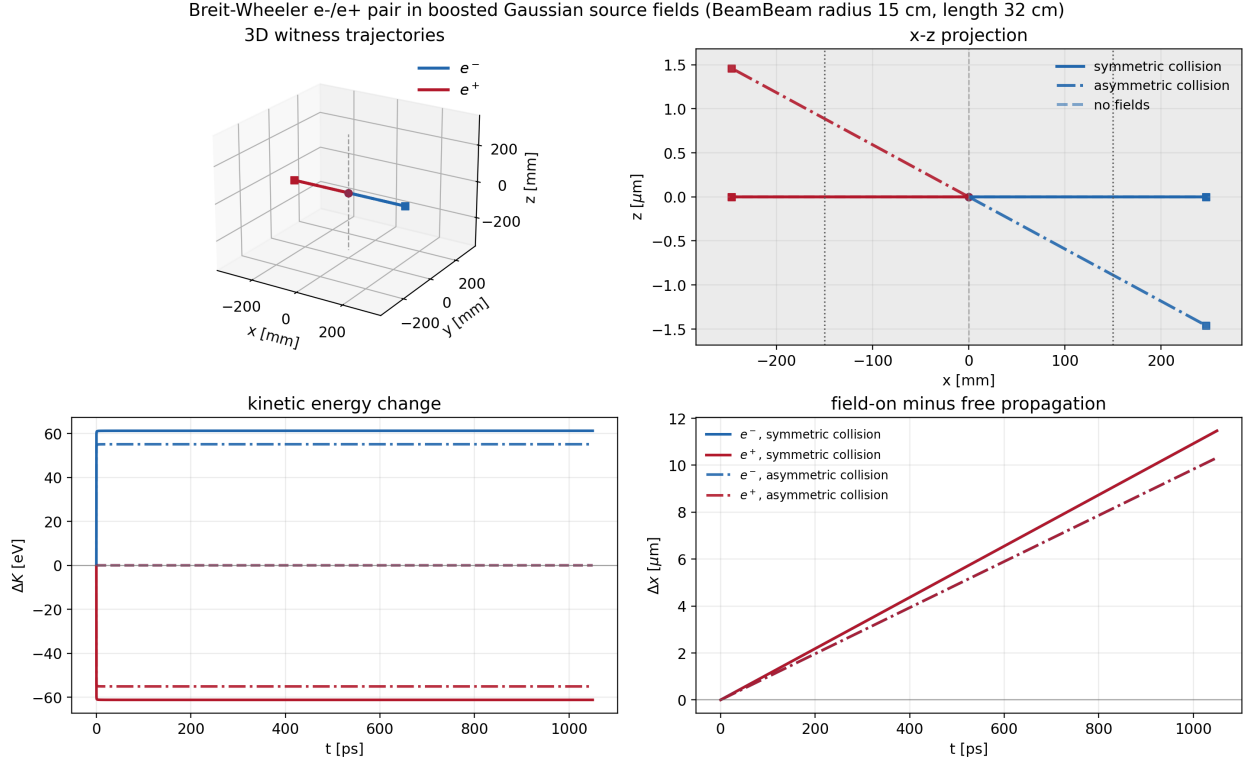


Figure 2: Back-to-back e^-/e^+ propagation with and without the colliding source fields over 1050 ps. The first 5 ps are integrated with 0.001 ps steps to resolve the overlap kick; the later drift uses 0.1 ps steps with 0.1 ps trajectory output. The x - z projection marks the $R_{\text{BB}} = 15$ cm cylinder scale and the 32 cm BeamBeam longitudinal window; the lower-right panel shows the field-on trajectory minus the free trajectory.

5. Only after this single-particle manufactured test passes should the implementation be generalized to many witnesses and mesh interpolation.

5 Sandbox Regression Overview

This section incorporates the BeamBeam sandbox regression overview into the manufactured solution note. The reusable regression and plotting scripts now live under `sandbox/python`; accepted metrics and figure copies used by this note live under `sandbox/note`.

5.1 Compared Cases

Check	Input / source	Comparison target
BeamBeam field dumps	<code>sandbox/BeamBeam-static-1V.in</code>	ASCII ρ , ϕ , and $\mathbf{E}$ dumps are compared against the analytic symmetric Gaussian manufactured solution.
Gamma-gamma e^-/e^+ tracking	<code>sandbox/track-e-p/gamma_gamma_pairs-2.in</code>	FROMFILE pair kinematics and final witness-container summaries are checked for the primary beam, electrons, and positrons.
OPALX-IMPACT drift	<code>sandbox/OPALX-IMPACT/drift-1.in</code>	A simple proton drift with space charge is compared with digitized IMPACT reference data.

5.2 Current Regression Metrics

Metric	Meaning	Current value
BeamBeam field snapshot	diagnostic kind	active beam-beam field diagnostics
charge integral	deposited charge	$-1.112650055 \times 10^{-14}$
ρ relative L_2	density error	5.2218586×10^{-2}
ϕ relative L_2	potential error	3.0422484×10^{-3}
E_x relative L_2	electric field error	1.5030245×10^{-2}
E_y relative L_2	electric field error	1.5821869×10^{-2}
E_z relative L_2	electric field error	2.4273686×10^{-2}
Metric	Meaning	Current value
OPALX rows	stat samples	73
IMPACT rows	digitized samples	71
final x	OPALX final centroid	3.4785129×10^{-2}
final ϵ_x	OPALX final emittance	1.8927107
RMS x difference	OPALX - IMPACT	7.6561747×10^{-4}
max x difference	OPALX - IMPACT	2.2219403×10^{-3}
RMS ϵ_x difference	OPALX - IMPACT	8.6413341×10^{-2}
max ϵ_x difference	OPALX - IMPACT	1.3101915×10^{-1}

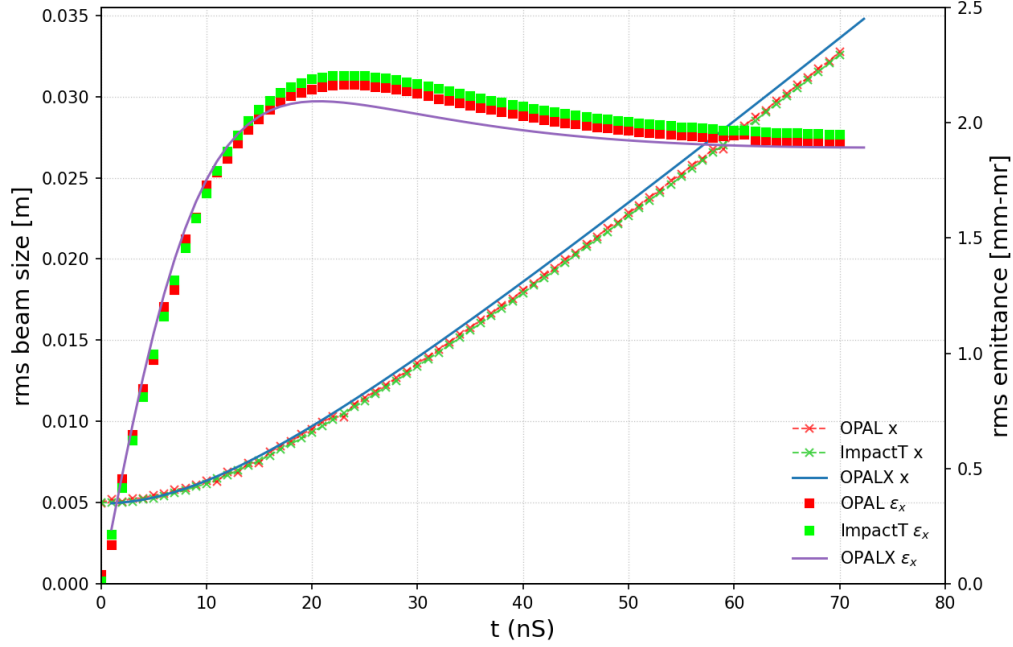


Figure 3: Digitized IMPACT reference data overlaid with the current OPALX drift result from `sandbox/OPALX-IMPACT/drift-1.stat`.

Metric	Meaning	Current value
electron count	FROMFILE particles	1297
positron count	FROMFILE particles	1297
electron mean K	input pair kinematics	313.601 keV
positron mean K	input pair kinematics	313.615 keV
median opening angle	pair summary	168.907°
opening > 150°	pair summary fraction	0.754819
electron final energy	gamma_gamma_pairs-2 c1	0.315272 MeV
positron final energy	gamma_gamma_pairs-2 c2	0.311939 MeV
electron final rms x	gamma_gamma_pairs-2 c1	1.3766180×10^{-2} m
positron final rms x	gamma_gamma_pairs-2 c2	1.3706767×10^{-2} m

5.3 BeamBeam Diagnostic Timeline

5.4 Cylinder Crossing Histograms

5.5 Figure Generation and Output Paths

The figure files used by this note are stored under `sandbox/note/figs` so the note does not depend on the regression output directory at compile time. The regression plotting scripts have been relocated to `sandbox/python` and default to note-local outputs. In particular, `make_sandbox_regression_overview.py` writes overview figures under `sandbox/note/figs`, `plot_cylinder_crossings.py` writes the staged large-cylinder histogram there, and `compare_cylinder_crossing_histograms.py` accepts `--output` for explicit figure placement.

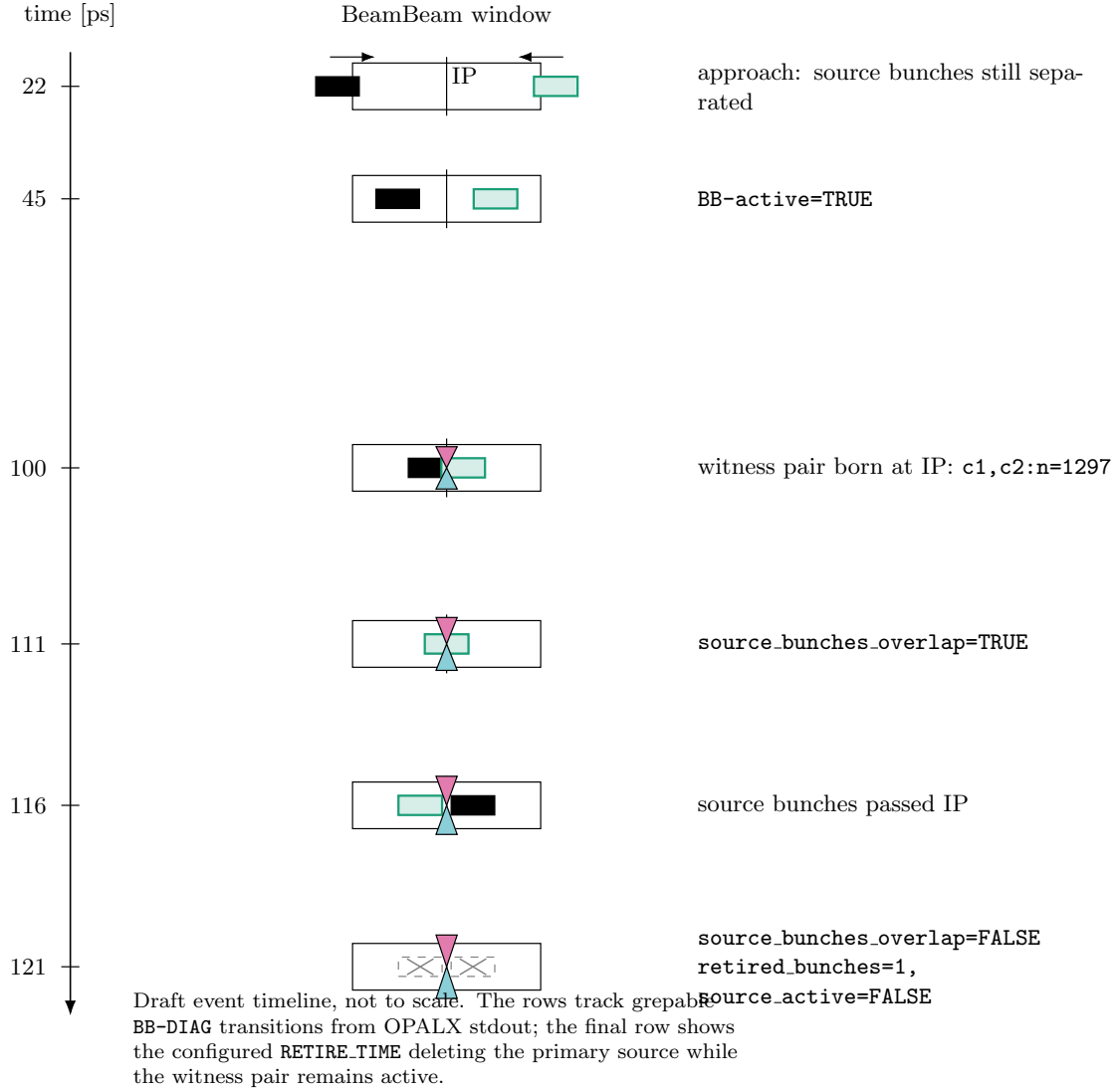


Figure 4: Timeline for copied BeamBeam source bunches and the injected low-energy witness pair.

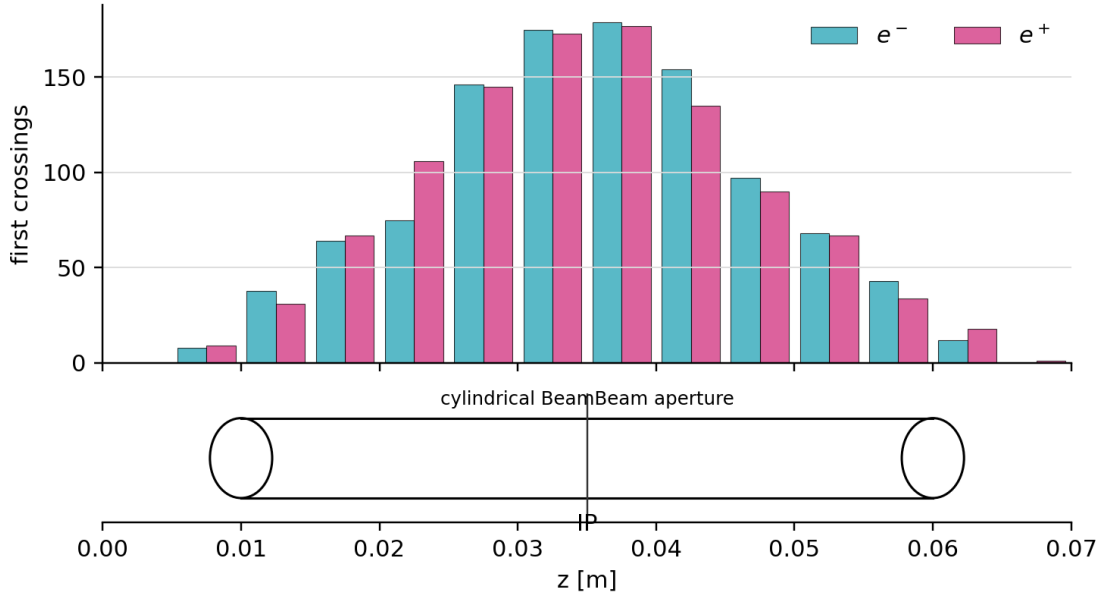
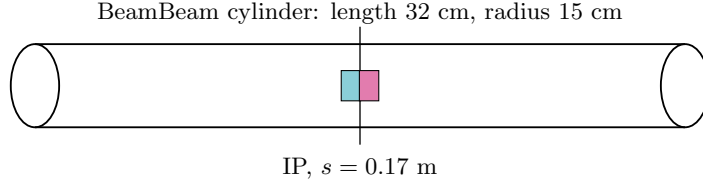


Figure 5: Cylinder-edge crossing histogram for the low-energy witness pair after source retirement. Bars show the first recorded H5 particle position with radius at least 1 cm for containers c1 and c2.



Event	Time	Note
BeamBeam window active	observed	BB-active=TRUE is emitted before witness injection.
Witness pair injected	550.3 ps	Preserves the 16.747 ps pre-IP offset from the proof-of-principle setup.
Primary reaches IP	567.1 ps	$s_{IP} = 0.17$ m for the 32 cm BeamBeam window.
Source retired	571.3 ps	Preserves the 4.253 ps post-IP retirement offset.
Tracking ends	1600 ps	Long enough for the first 15 cm radial crossings.

Figure 6: Timing and geometry summary for the large-cylinder setup.

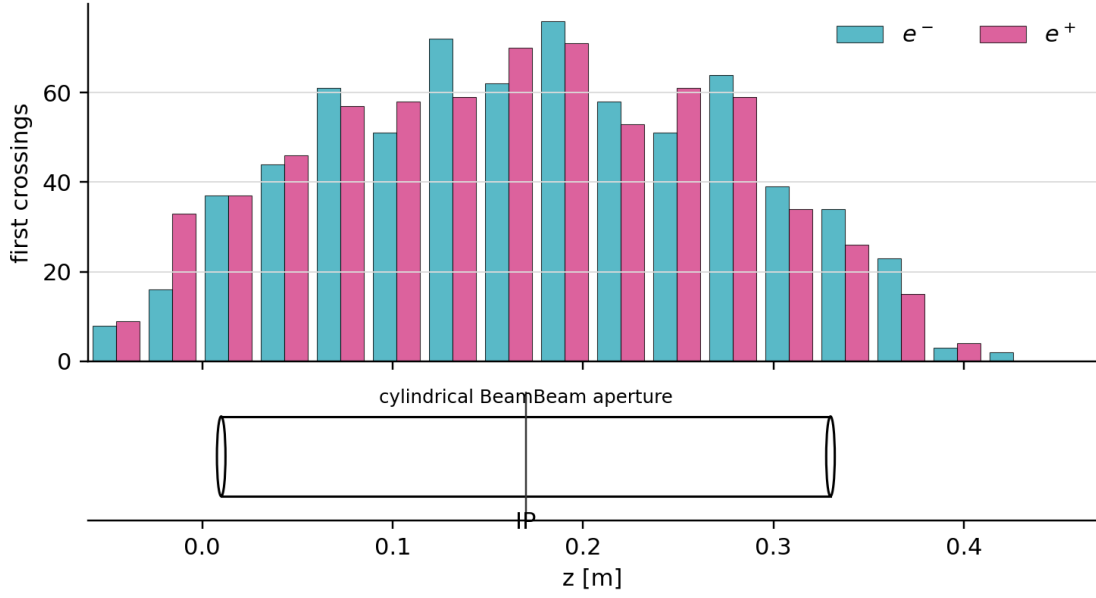


Figure 7: Cylinder-edge crossing histogram for the large setup with 15 cm radius and 32 cm BeamBeam length. Bars show first recorded H5 particle positions with radius at least 15 cm for containers c1 and c2.

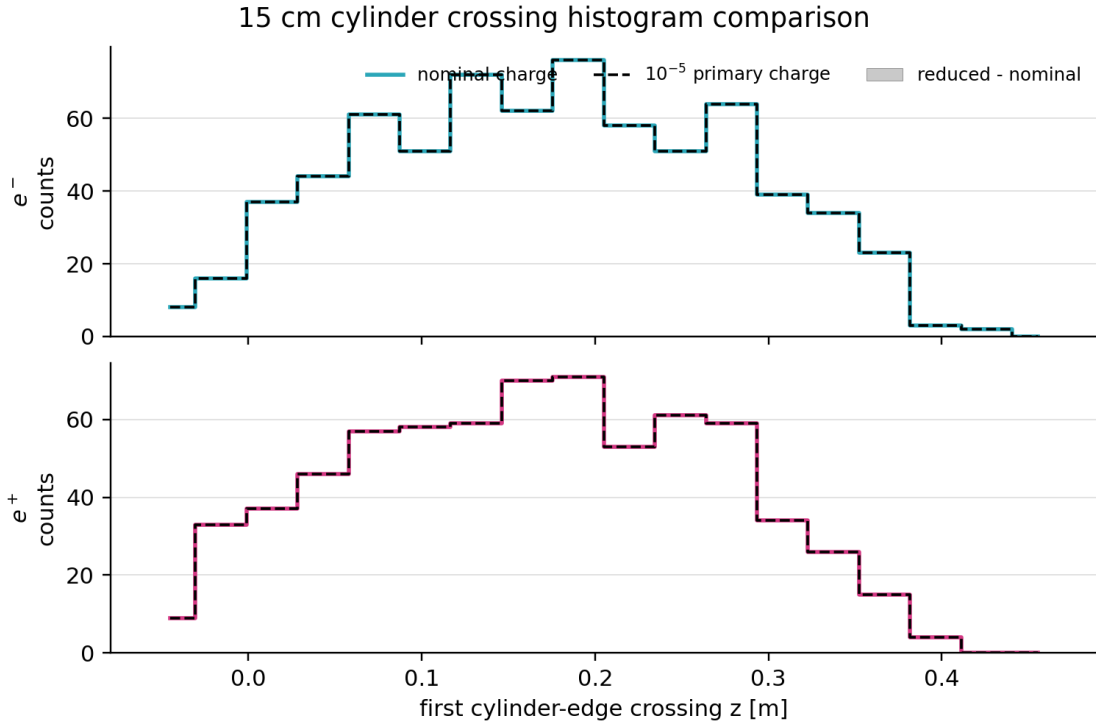


Figure 8: Comparison of nominal large-cylinder crossing histograms with a rerun where the primary bunch charge is scaled by 10^{-5} . Solid curves show the nominal primary charge, dashed curves show the reduced-charge run, and grey bars show the reduced-minus-nominal bin difference.

5.6 Regression Workflow Notes

The lightweight regression checks are now driven from `sandbox/python/run_sandbox_regressions.py`. They cover three compact checks: manufactured `BeamBeam` field dumps, gamma-gamma e^-/e^+ FROMFILE pair tracking, and the OPALX-IMPACT drift comparison. The accepted baseline is the JSON file `sandbox/note/sandbox_regression_baseline.json`; the latest flattened metrics are written to `sandbox/note/current_metrics.csv`. The CSV is run output, while the JSON file is the comparison reference.

The baseline should only be updated after an intentional physics or numerics change. When accepting new results, record the expected impact on conservation, stability, floating-point reproducibility, and OpenMP/MPI execution order. The baseline JSON records the git branch, commit, dirty flag, and `git status --short` output present when the baseline was accepted.

The pair-input utilities remain in `sandbox/track-e-p`. The standard six-column FROMFILE distributions are regenerated with `convert_fort98_to_fromfile.py`. Creation-time diagnostics use the `--add-time` option, which writes `x y z px py pz t bin_number` with $t = ct/c$ in seconds and `bin_number=1`. The analyzer `analyze_pair_momenta.py` reports the corresponding time summaries in picoseconds and can write the time histogram diagnostics.

5.7 Staged Timestep Rationale

OPALX interprets vector-valued `DT` and `MAXSTEPS` entries as per-segment tracking instructions. The entries are not cumulative step numbers. For example,

```
MAXSTEPS = {550, 25001, 1025}
DT        = {1.0e-12, 1.0e-15, 1.0e-12}
```

means 550 coarse 1 ps steps, followed by 25001 fine 1 fs steps, followed by 1025 coarse 1 ps steps. The total simulated time is therefore

$$550 \text{ ps} + 25001 \text{ fs} + 1025 \text{ ps} = 1600.001 \text{ ps}. \quad (5.1)$$

The fine segment starts at 550 ps. This deliberately brackets the proof-of-principle witness injection time

$$t_{\text{inj}} = t_{\text{IP}} - 16.747 \text{ ps} \simeq 550.312 \text{ ps}, \quad (5.2)$$

and continues through the primary bunch IP crossing at

$$t_{\text{IP}} = \frac{0.17 \text{ m}}{c} \simeq 567.059 \text{ ps}. \quad (5.3)$$

The reason is operational as well as numerical: the FROMFILE source is a one-shot delayed emitter. It emits when the active time interval $[t, t + \Delta t)$ crosses `T0`; it does not itself request a smaller emission timestep. Thus the global tracker timestep must already be fine before t_{inj} if pair creation is to occur with femtosecond resolution.

For the histogram-focused production run, only the witness trajectory samples are needed to find the first cylinder-edge crossing,

$$\sqrt{x^2 + y^2} \geq 0.15 \text{ m}. \quad (5.4)$$

The integration still uses the 1 fs segment above, but dense diagnostic output is not required. The staged input therefore uses

```
OPTION, PSDUMPFREQ = 10;  
OPTION, STATDUMPFREQ = 0;
```

so H5 phase-space samples are written every ten integration steps and stat files are suppressed. In the coarse tail this gives approximately 10 ps sampling, corresponding to about 3 mm of light-travel distance, which is small compared with the current histogram span. The source-field ASCII `.dat` files produced by the debug-enabled field solver are not used for the histogram and can be deleted during or after the run.

5.8 Reproduction Commands

```
source .venv-h6/bin/activate  
python sandbox/python/run_sandbox_regressions.py \  
    --run-opalx \  
    --opalx-exe build_openmp/src/opalx  
cd sandbox/track-e-p  
../../build_openmp/src/opalx gamma_gamma_pairs-large-cylinder.in  
../../build_openmp/src/opalx gamma_gamma_pairs-large-cylinder-q1em5.in  
cd ../../..  
python sandbox/python/make_sandbox_regression_overview.py  
cd sandbox/note  
latexmk -pdf boosted_gaussian_witness.tex
```